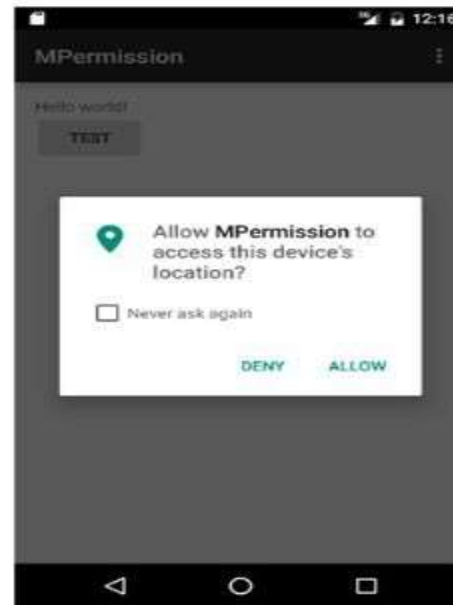


Android Permission

Android Permission

- In Android 6.0 Marshmallow, application will not be granted any permission at installation time. Instead, application has to ask user for a permission one-by-one at runtime.



Normal Permissions

```
<!--Normal Permissions-->  
<uses-permission android:name="android.permission.INTERNET"/>  
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE"/>  
<uses-permission android:name="android.permission.BLUETOOTH"/>  
<uses-permission android:name="android.permission.CHANGE_NETWORK_STATE"/>  
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE"/>
```

Dangerous Permissions

```
<!--Normal Permissions-->
<uses-permission android:name="android.permission.INTERNET" />
<uses-permission android:name="android.permission.CHANGE_WIFI_STATE" />

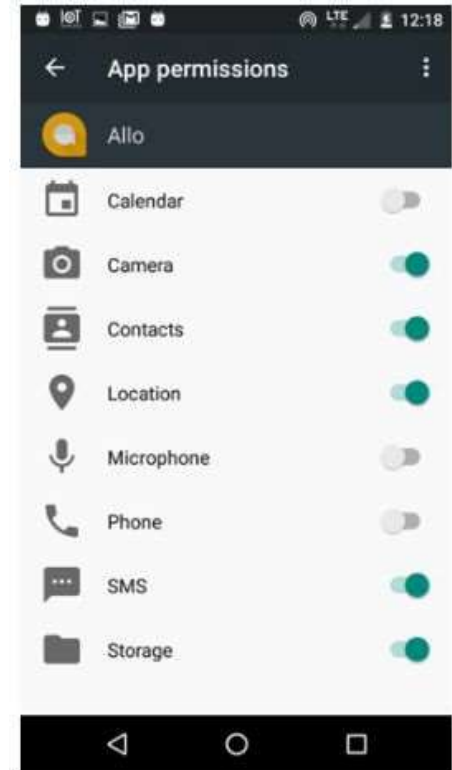
<!--Dangerous Permission-->
<uses-permission android:name="com.google.android.providers.gsf.permission.READ_GSERVICES" />
<uses-permission android:name="android.permission.CAMERA" />
<uses-permission android:name="android.permission.CALL_PHONE" />
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.SEND_SMS" />
<uses-permission android:name="android.permission.RECEIVE_SMS" />
<uses-permission android:name="android.permission.GET_ACCOUNTS" />
<uses-permission android:name="android.permission.READ_PHONE_STATE" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
```

Permission Groups

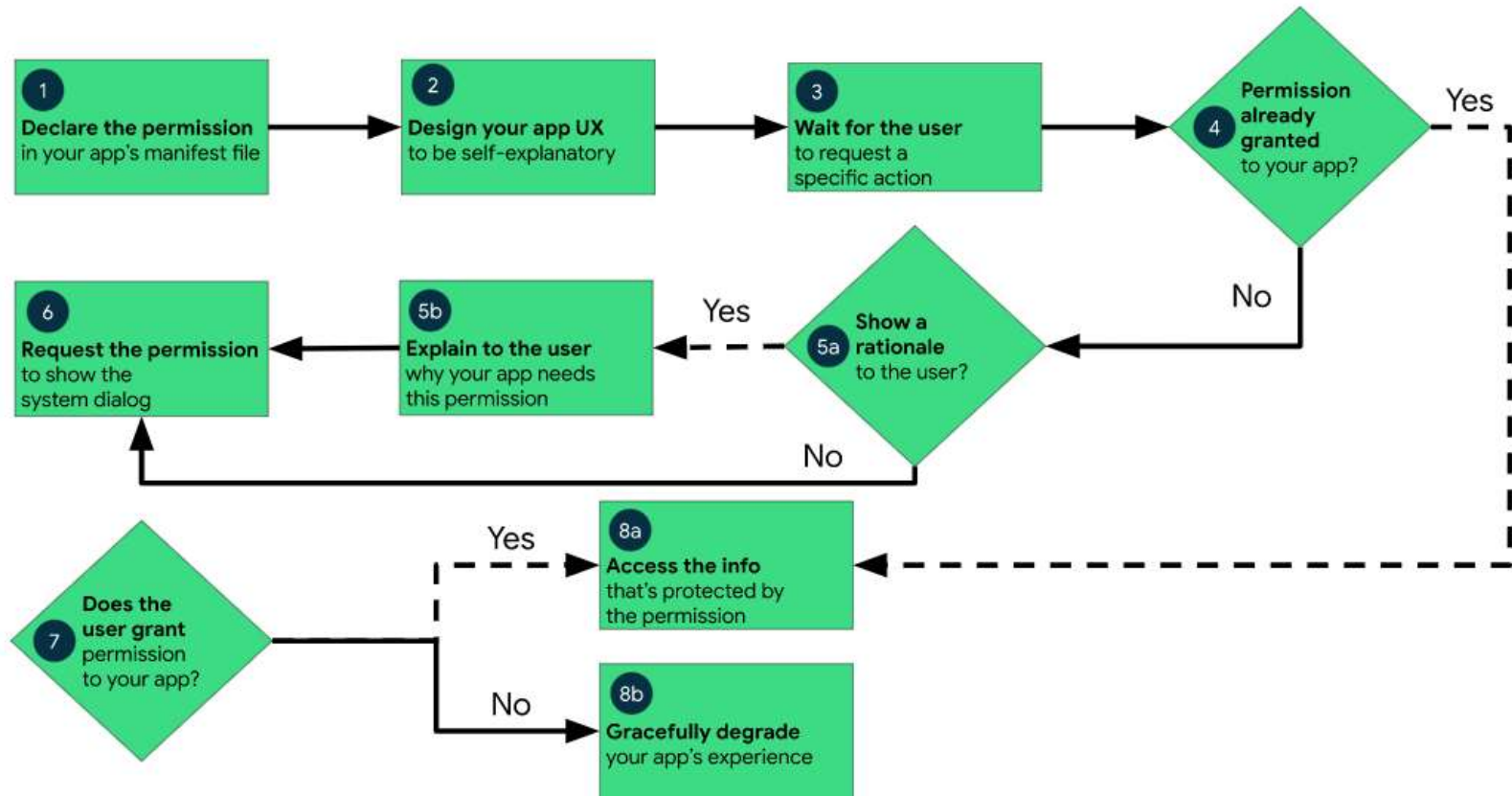
- **Android 6.0 Marshmallow has nine main groups of permissions:**
 - Calendar: Read and/or write to the calendar.
 - Camera: Give the application the ability to access the camera. Location: Access fine or coarse location.
 - Microphone: The ability to record audio.
 - Phone: Includes phone state; the ability to make calls, read, and write to the call log; and voicemail. Sensor: The ability to use various sensors in the device, like a gyroscope.
 - SMS: Similar to how the phone is handled including sending and receiving texts. MMS and cell broadcasts. Storage: Read and write to device's external storage.

Manually Enabling and Disabling Runtime Permissions

- **Open your device's Settings app.**
 - Tap apps and find an app that you want to work with. Select it.
 - Tap applications permissions on the App Info screen.
 - You will see the list of permissions the app requests. Tap on the switch to turn it on or off.



Workflow for handling android permission



How to Check for Runtime Permissions

- Check the Target SDK Version

```
if ((Build.VERSION.SDK_INT >= Build.VERSION_CODES.M)) {  
    getLocationPermission();  
}
```

- Check if Permission Is Granted

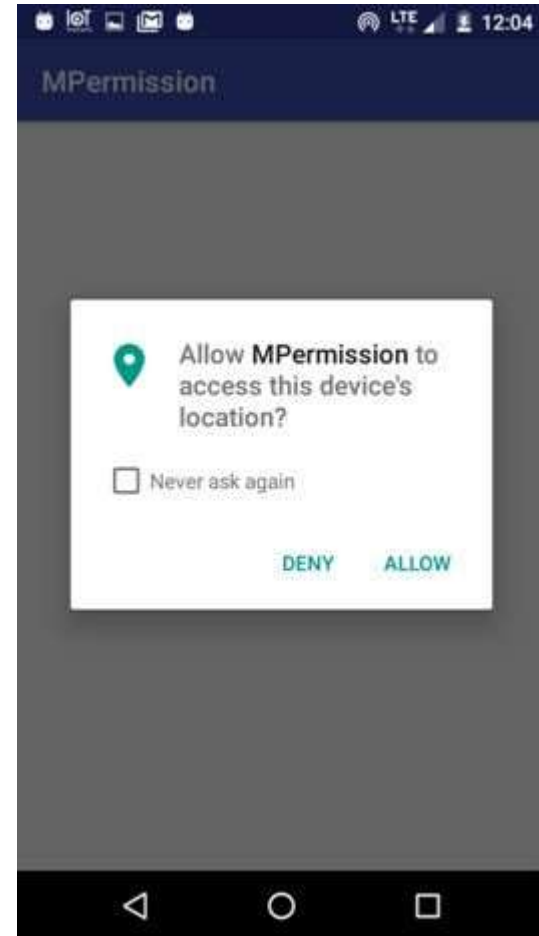
```
if ((ContextCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION) != PackageManager.PERMISSION_GRANTED)) {  
    /*Ask for permission*/  
}
```

Handle "Never Ask Again"

```
if (shouldShowRequestPermissionRationale(  
    Manifest.permission.ACCESS_FINE_LOCATION)) {  
  
    //Explain to the user why we need to location service.  
}
```

Request for Permission

```
if ((ContextCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION) != PackageManager.PERMISSION_GRANTED)) {  
    /*Ask for permission*/  
  
    //need to request permission  
    if (shouldShowRequestPermissionRationale(  
        Manifest.permission.ACCESS_FINE_LOCATION)) {  
        //Explain to the user why we need to location service.  
    }  
  
    /*Request for permission*/  
    requestPermissions(new String[]{Manifest.permission.ACCESS_FINE_LOCATION},  
        REQUEST_CODE_FOR_SOURCE);  
}
```



Handle the Permissions Request Response

```
@Override
public void onRequestPermissionsResult(int requestCode,
                                     String permissions[], int[] grantResults) {
    switch (requestCode) {
        case REQUEST_CODE_FOR_SOURCE: {
            // If request is cancelled, the result arrays are empty.
            if (grantResults.length > 0
                && grantResults[0] == PackageManager.PERMISSION_GRANTED) {

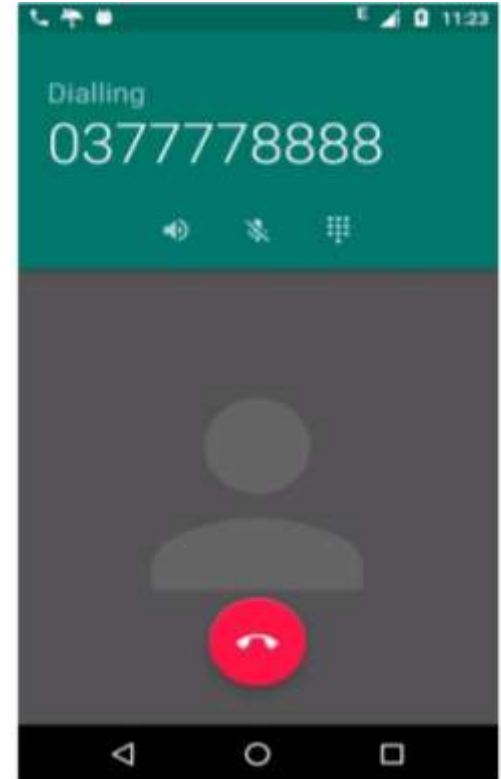
                // permission was granted, yay! Do the
                // location related task you need to do.

            } else {

            }
        }
        return;
    }
}
```

Classwork

- In Product Details screen, add phone number of Store
- Click Phone number, call the Store



References

- <https://developer.android.com/training/permissions/requesting>
- <https://dzone.com/articles/android-runtime-permissions>
- [**https://inthecheesefactory.com/blog/things-you-need-to-know-about-android-m-permission-development/en**](https://inthecheesefactory.com/blog/things-you-need-to-know-about-android-m-permission-development/en)